

Input files

File name	Content
param.dat	Parameters controlling TRACMT. Lines prefixed with # are recognized as comment lines.
<i>Arbitrary names</i>	Time-series data

Output files

File name	Content
TRACMT.log	Execution logs
TRACMT.cvg	Detained information of computation processes
response_functions.csv	Resultant transfer function estimates and their error estimates
apparent_resistivity_and_phase.csv	Estimates of the apparent resistivity and phase and their error estimates (keyword “OUTPUT_RHOA_PHS” is required)

How to run TRACMT

You need to execute the following command in the directory where input files exist.

TRACMT

File format of param.dat (1/7)

Keyword	Content	Data type	Option	Default	Example
NUM_OUT	Number of output channels ^{*1)}	Positive Integer			NUM_OUT 2
SAMPLING_FREQ	Sampling frequency (Hz)	Positive real number			SAMPLING_FREQ 32
NUM_THREADS	Number of OpenMP threads	Positive integer			NUM_THREADS 4
AZIMUTH	Measurement direction (clockwise angle (deg.) from the north) of all channels including remote reference data	Real numbers			AZIMUTH 0.0 90.0 0.0 90.0 0.0 90.0
ROTATION	Rotation angle (deg.) (positive clockwise) of the coordinate system.	Real number		0	ROTATION 30
CAL_FILES	File names used for correcting instrumental calibration ^{*1)}	Characters		Calibration is not performed	CAL_FILES hz.cal hx.cal hy.cal hrx.cal hry.cal

*1) The file format of the correction file is shown in a subsequent slide.

File format of param.dat (2/7)

Keyword	Content	Data type	Option	Default	Example
PROCEDURE	Transfer function estimation method	Integer	0: Ordinary robust remote reference method 1: RRMS estimator*1)	0	PROCEDURE 0
RRMS	Parameters of RRMS estimator	Detail of this option is described in a subsequent slide		1 100 3 0.05 10 16 0.01	RRMS 1 100 3 0.05 10 16 0.01
MESTIMATORS*2)	M-estimators used in the ordinary robust remote reference method*3)	Two integers for the first and second M-estimators		0 1	MESTIMATORS 0 1

*1) Remote reference data are necessary when the RRMS estimator is used.

*2) To use the fast and robust bootstrap method, it is necessary to write the option as “1 -1”.

*3) In the ordinary robust remote reference method, iterative re-weighted least square (IRWLS) is performed twice. In the first IRWLS, the scale of the residual magnitude is also updated as well as transfer functions, while the scale parameter is fixed in the second IRWLS.

If you write “-1 -1” in this option, the ordinary least square is performed.

*4) The severe weight function used in Chave & Thomson (1989, 2004)

File format of param.dat (3/7)

Keyword	Content	Data type	Option	Default	Example
ERROR_ESTIMATION	Error estimation method	Integer	0: Parametric approach 1: Fixed-weight bootstrap 2: Ordinary (strict) bootstrap 3: Fast and robust bootstrap 4: Fixed-weight Jackknife 5: Subset deletion Jackknife (Eisel & Egbert, 2001)	1	ERROR_ESTIMATION 1
BOOTSTRAP	Number of bootstrap samples	Positive integer		1000	BOOTSTRAP 1000
PREWHITENING	Detail of this option is described in a subsequent slide	Three integers		Any pre-whitening is NOT performed	PREWHITENING 0 10 5
ROBUST_FILTER ^{*1)}	Detail of this option is described in a subsequent slide	Integers		Robust filter is not applied	ROBUST_FILTER 0 10 12 50 10 12 50 10 12 50 10 12 50 10 12 50 10 12 50

*1) The robust filter is NOT recommended because fine parameter tuning is required.

File format of param.dat (4/7)

Keyword	Content	Data type	Option	Default	Example
DECIMATION	Decimation interval, length of antialiasing (FIR) filter, and width between the passband and stopband* ¹⁾ frequency in the logarithmic scale.	Two positive integers and one positive real number		Decimation (down-sampling) is not performed.	DECIMATION 32 100 0.5
HIGH_PASS	Cutoff frequency of high-pass filter (Hz)	Positive real number		HPF is not applied	HIGH_PASS 0.1
LOW_PASS	Cutoff frequency of low-pass filter (Hz)	Positive real number		LPF is not applied	LOW_PASS 0.1
NOTCH	Cutoff frequencies (Hz) following the number of them	Integer and positive real numbers		Notch filters are not applied	NOTCH 6 50 100 150 200 250 300
COHERENCE_CRITERIA	Number of segments used for calculating transfer functions followed by the threshold of squared coherence* ¹⁾	Positive integer and non-negative real number		Coherence thresholding is not performed.	COHERENCE_CRITERIA 10 0.3

*1) The stopband frequency is the Nyquist frequency after decimation.

*2) If the squared coherence is lower than the threshold, all segments used for calculating the corresponding transfer functions are excluded in the transfer function estimation.

File format of param.dat (5/7)

Keyword	Content	Data type	Option	Default	Example
ATS_BINARY	Read ats files of Metronix instruments				ATS_BINARY
MFS_CAL	Coil carburation files of Metronix instruments ^{*2)} .	Characters and/or real numbers		Calibration is not performed	MFS_CAL 30.0 30.0 MFS06375.TXT MFS06376.TXT MFS06e549.TXT MFS06e576.TXT
OUTPUT_TIME_SERIES	Output input time-series data to csv files				OUTPUT_TIME_SERIES
OUTPUT_RHOA_PHS	Output apparent resistivity and phase to a csv file in addition to the impedance tensor ^{*3)}				OUTPUT_RHOA_PHS
LOGGER_CAL_DIRECTORY	Path of the directory in which the filter coefficient files are located	Character		Current directory	LOGGER_CAL_DIRECTORY ../Calibraion_Files

*1) The file format of the calibration files are shown in a subsequent slide. Carburation files for the remote reference data are also required.

*2) If you write a real number instead of a character, this number is recognized as the dipole length (m) of an electric field channel.

*3) Apparent resistivity (Ωm) is calculated assuming that the units of the electric and magnetic field is mV/km and nT, respectively.

File format of param.dat (6/7)

Keyword	Content	Data type	Option	Default	Example
ELOGMT_BINARY	Read dat files of ELOG-MT				ELOGMT_BINARY
ELOGMT_READ_OPTION	Components that are read from dat files of ELOG-MT	Integer	0: $E_x, E_y, H_z, H_x, H_y, H_{rx}, H_{ry}$ 1: $E_x, E_y, H_x, H_y, H_{rx}, H_{ry}$ 2: $H_z, H_x, H_y, H_{rx}, H_{ry}$ 3: E_x, E_y, H_z, H_x, H_y 4: E_x, E_y, H_x, H_y 5: H_z, H_x, H_y 6: E_x, E_y 7: H_x, H_y, H_{rx}, H_{ry} 8: H_x, H_y	0	ELOGMT_READ_OPTION 0
ELOGMT_CAL	Options used for the calibration of ELOG-MT	Integer Character Real number			ELOGMT_CAL 0 ../Calib/ELOGMT_SN4042.txt 0.0
ELOGDUAL_BINARY	Read dat files of ELOG-DUAL or ELOG1K				ELOGDUAL_BINARY
ELOGDUAL_CAL	Options used for the calibration of ELOG-DUAL or ELOG1K	Integer Character Real number			ELOGDUAL_CAL 0 ../Calib/ELOG_No10.txt 0.0

File format of param.dat (7/7)

Keyword	Content	Data type	Option	Default	Example
MTH5	Read MTH5 files (Peacock et al., 2022)				MTH5
MTH5_FILTERS	Read filters from MTH5 files.				MTH5_FILTERS
NUM_SECTION	Number of time-series data files for each channel	Positive Integer			NUM_SECTION 1
SEGMENT	Segments lengths (section lengths) of the overlapped section averaging method and indexes of the Fourier transforms ^{*2)}	Positive integers (detail of this option is described in a subsequent slide)			SEGMENT 3 1024 3 2 3 4 512 2 3 4 256 2 3 4
DATA_FILES	File names of time-series data and numbers of data to be read	Detail of this option is described in a subsequent slide			Detail of this option is described in the next slide
MERGE_SECTIONS	Merge time-series of different files into a continuous time-series. (detail of this option is described in a subsequent slide)	Integers.		Merge operation is not performed	MERGE_SECTIONS 2 0 2 3 5
END	End of controlling parameters				END

Keyword RRMS

RRMS

Are initial candidate sets of the ISTFs randomly selected for each frequency?

0: No. TFs at the previous (lower) frequency are used as an initial candidate set.

1: Yes.

Number of the initial candidate sets of the ISTFs (positive integer)

Maximum number of iteration of the first parameter updates (positive integer)

Convegence criteria of the first parameter updates (positive real number)

Maximum number of the candidate sets of the second parameter updates (positive integer)

Maximum number of iteration of the second parameter updates (positive integer)

Convegence criteria of the second parameter updates (positive real number)

Default setting:

RRMS

1

100

3

0.05

10

16

0.01

Convegence criteria means the threshold of the change rates of weighted norm of each residual and the robust scale of the Mahalanobis distance.

Keyword PREWHITENING

PREWHITENING

Prewhitening method

-1: Pre-computed AR coefficients are specified

0: Non-robust prewhitening

1: Robust prewhitening

Maximum degrees of the AR model used for prewhitening¹⁾

Number of initial candidates (it is used only in the robust prewhitening)

If you do not write this keyword, any pre-whitening is not performed

1) I recommend you use a higher value than five.

Keyword ROBUST_FILTER

ROBUST_FILTER

Are filtered time-series directly used for the transfer function estimation

0: No

1: Yes

Parameters for the 1st channel

The 1st threshold

The 2nd threshold

Maximum number of
consecutive replacements

Parameters for the 2nd channel

The 1st threshold

The 2nd threshold

Maximum number of
consecutive replacements

⋮

NOTE:
The robust filter is NOT recommended
because fine parameter tuning is required.

Number of
data channels

Keyword SEGMENT

SEGMENT

Number of segment lengths

The 1st segment length

Number of indexes

The 1st index

...

The last index

The 2nd segment length

Number of indexes

The 1st index

...

The last index

⋮

Number of indexes

Number of
segment lengths

[NOTE]

- Segment length should be a power of two.
- I recommend you use three or higher as indexes if the prewhitening is performed and otherwise use seven or higher.

Frequency (Hz) for which the transfer function is estimated is determined from the segment length (L), index (i), and sampling interval (Δt), as follows.

$$f = \frac{i}{L\Delta t}$$

For example, the following setting leads to the transfer functions at 0.2344 and 0.3125 (Hz) if the sampling interval is 0.1 (sec).

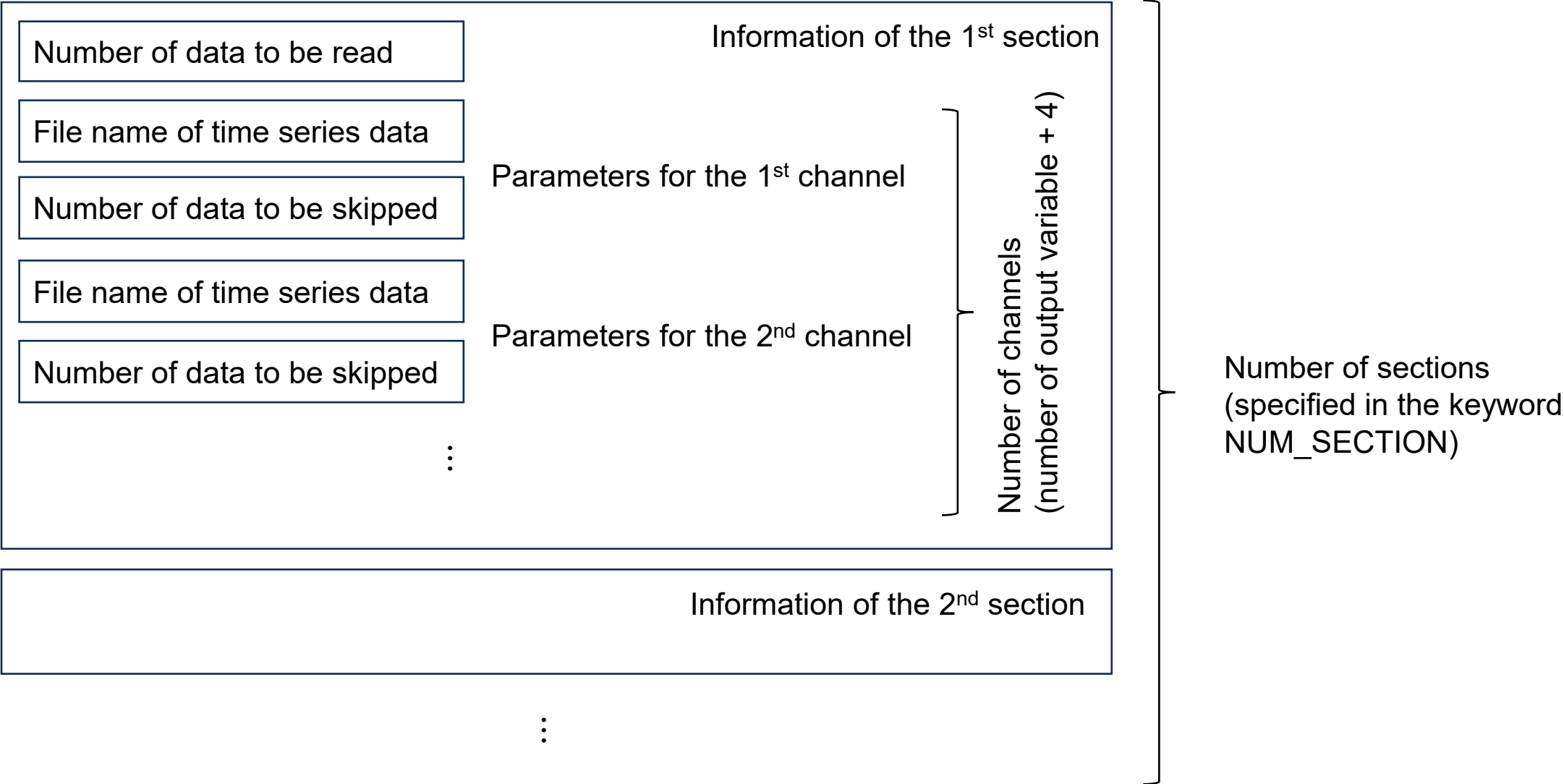
Segment length	Number of indexes	The 1 st index	The 2 nd index
128	2	3	4

$$f_1 = \frac{3}{128 \times 0.1} = 0.2344 \text{ (Hz)}$$

$$f_2 = \frac{4}{128 \times 0.1} = 0.3125 \text{ (Hz)}$$

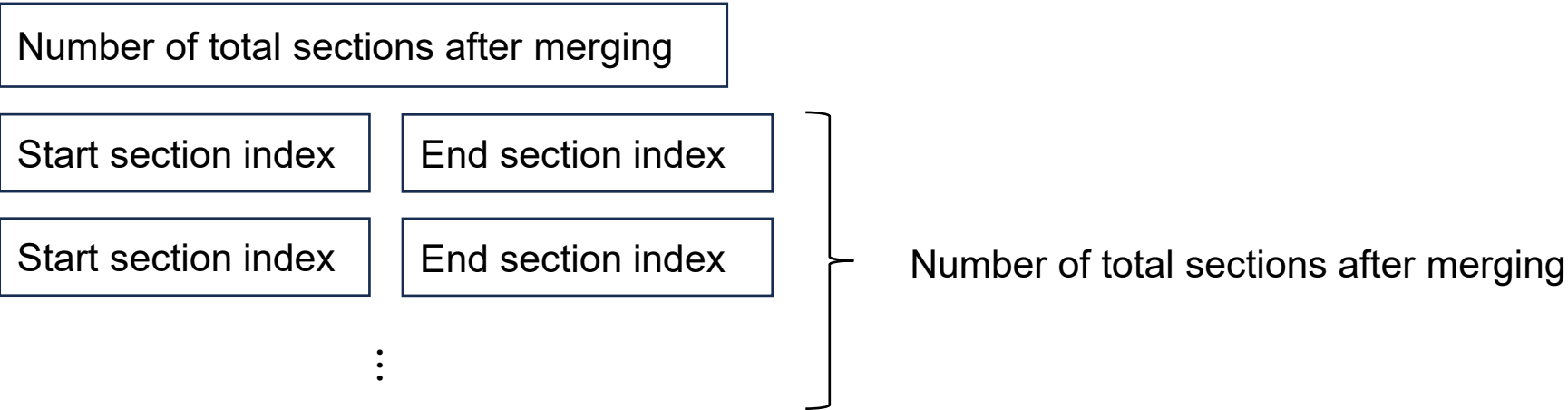
Keyword DATA_FILES

DATA_FILES



Keyword MERGE_SECTIONS

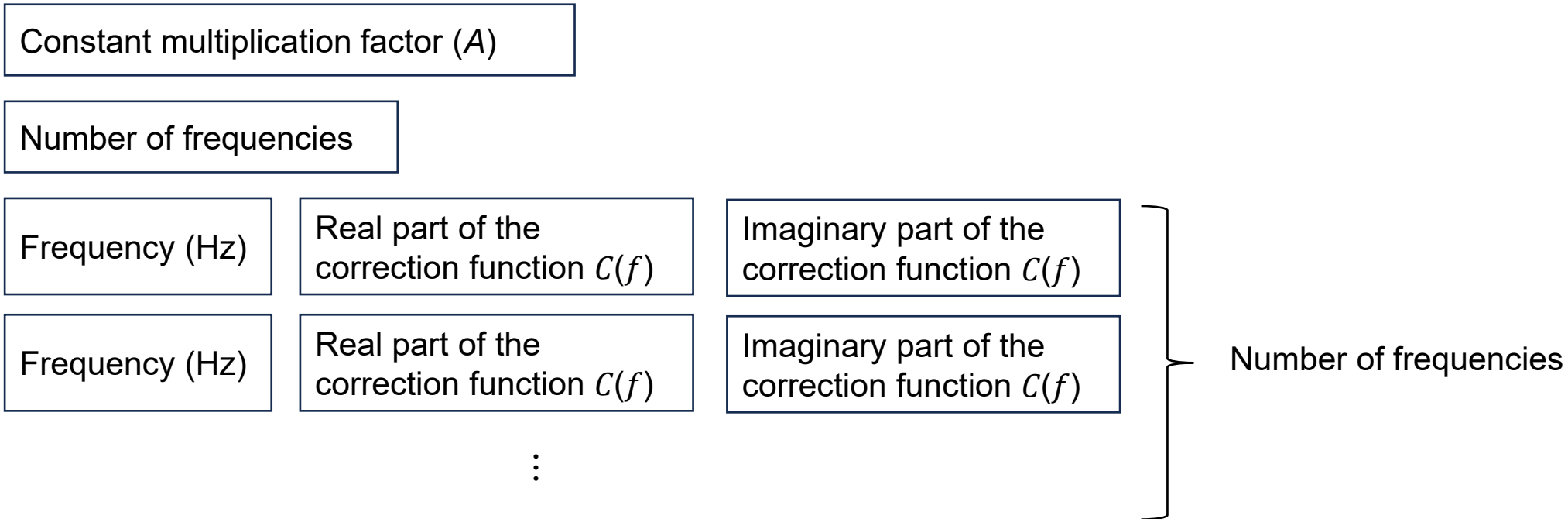
MERGE_SECTIONS



For example, following setting merges the data from the section #0 to the section #3 into a section without gap.

```
MERGE_SECTIONS
1
0 3
```


Format of the files for correcting instrumental characteristics



Instrumental characteristics are corrected in the frequency domain by multiplying A and $C(f)$.

The correction function is interpolated for the frequency at which $C(f)$ is not specified.

$$A \times C(f_1)E_x(f_1) \rightarrow E_x(f_1)$$

$$A \times C(f_2)E_x(f_2) \rightarrow E_x(f_2)$$

...

File format of output file

The resultant transfer functions are outputted to csv files with headers such as “resp_real_0_2”, “resp_imag_0_2”, “coherence_0_2+3”, and “dresp_0_2”.

The number of header indicates index of channels. For example,

		Channel indexes							
		0	1	2	3	4	5	6	
Variables		E_x	E_y	H_z	H_x	H_y	H_{rx}	H_{ry}	(Number of output variables is 3)
		E_x	E_y	H_x	H_y	H_{rx}	H_{ry}		(Number of output variables is 2)
		H_z	H_x	H_y	H_{rx}	H_{ry}			(Number of output variables is 1)

- resp_real_i_j: Real parts of the transfer function relating channel #i to channel #j
- resp_imag_i_j : Imaginary parts of the transfer function relating channel #i to channel #j
- coherence_i: Squared coherence for channel#i
- dresp_i_j: Standard error for the transfer function relating channel #i to channel #j

For example, if the output variables are Ex and Ey, resp_real_0_3 and resp_imag_0_3 are real and imaginary components of the Z_{xy} component of the impedance tensor.

$$\begin{matrix} & & & 0_2 & 0_3 \\ & & & & \\ 0 & \left(\begin{matrix} E_x \\ E_y \end{matrix} \right) & = & \left(\begin{matrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{matrix} \right) & \left(\begin{matrix} H_x \\ H_y \end{matrix} \right) & \begin{matrix} 2 \\ 3 \end{matrix} \\ & & & 1_2 & 1_3 \end{matrix}$$

How to read ats files of Metronix instruments

ATS_BINARY

MFS_CAL

30.0

30.0

MFS06375.TXT

MFS06376.TXT

MFS06e549.TXT

MFS06e576.TXT

DATA_FILES

691200

063_V01_C00_R008_TEx_BL_32H.ats

0

063_V01_C01_R008_TEy_BL_32H.ats

0

063_V01_C02_R008_THx_BL_32H.ats

0

063_V01_C03_R008_THy_BL_32H.ats

0

382_V01_C02_R069_THx_BL_32H.ats

0

382_V01_C03_R069_THy_BL_32H.ats

0

If you write a real number instead of a character, this number is recognized as a dipole length (km) of an electric field channel, and the data at the channel is divided by the real number in the calibration stage.

TRACMT can read ats files when the keyword “ATS_BINARY” is written.

File extensions should be ‘ats’.

How to read MTH5 files

MTH5

MTH5_FILTERS Filter information is read from MTH files.

DATA_FILES

750000

sample1.mth5 /schedule_01/ex

0

sample1.mth5 /schedule_01/ey

0

sample1.mth5 /schedule_01/hx

0

sample1.mth5 /schedule_01/hy

0

sample2.mth5 /schedule_01/hx

150000

sample2.mth5 /schedule_01/hy

150000

Names of MTH5 files

Group names of each channel (channel path)

How to read dat files of ELOG-MT

ELOGMT_BINARY

ELOGMT_READ_OPTION

1

- | | | |
|--|------------------------------|-------------------------------|
| 0: $E_x, E_y, H_z, H_x, H_y, H_{rx}, H_{ry}$ | 3: E_x, E_y, H_z, H_x, H_y | 6: E_x, E_y |
| 1: $E_x, E_y, H_x, H_y, H_{rx}, H_{ry}$ | 4: E_x, E_y, H_x, H_y | 7: H_x, H_y, H_{rx}, H_{ry} |
| 2: $H_z, H_x, H_y, H_{rx}, H_{ry}$ | 5: H_z, H_x, H_y | 8: H_x, H_y |

ELOGMT_CAL

0

→ 0 (ADU mode) or 1 (PHX mode)

../Calib/ELOGMT_SN4042.txt → ELOG-MT calibration file with its path. The file format is shown in the next slide.

0.0

→ A real number used to correct the time difference between loggers.

LOGGER_CAL_DIRECTORY

When all data are measured with ELOG, one can write 0.0 because no correction is required.

../../Calib

→ Under this directory, the filter coefficient files of ELOG-MT should be located.

(continue to the next slide)

- ADU mode: firh_adu.txt, fir_l_adu.txt
PHX mode: firh_phx.txt, fir_l_phx.txt, fir_r_phx.txt

How to read dat files of ELOG-MT

DATA_FILES

115200

D:Site1/20221023/20221023-010000_32Hz.dat

0

D:Site1/20221023/20221023-010000_32Hz.dat

0

D:Site1/20221023/20221023-010000_32Hz.dat

0

D:Site1/20221023/20221023-010000_32Hz.dat

0

D:Site2/20221023/20221023-010000_32Hz.dat

0

D:Site2/20221023/20221023-010000_32Hz.dat

0

TRACMT can read .dat files of ELOG-MT when the keyword “ELOGMT_BINARY” is written.

If your compiler supports C++17, you can use the following function by activating the preprocessor “_USE_FILESYSTEM” when compiling.

If you write “*.dat” under the directory name, all data files (of corresponding sampling frequency) under the directory are automatically read.

The file format of the ELOG-MT calibration file

- The first line is the comment line.
- Below the first line, frequency-dependent characteristics of each channel should be written, comma delimited.
- The first column is the frequency (Hz). The second and third columns are the amplitudes and phase (deg.) of Ch#1 (E_x), respectively. The fourth and fifth columns are the amplitudes and phase (deg.) of Ch#2 (E_y), respectively. In the following columns, the amplitudes and phase (deg.) of the magnetic field channel should be written in the same way.

Example of the ELOG-MT calibration file

	freq(Hz)	CH1 (mVpp)	CH1 (deg)	CH2 (mVpp)	CH2 (deg)	CH3 (mVpp)	CH3 (deg)	CH4 (mVpp)	CH4 (deg)	CH5 (mVpp)	CH5 (deg)
1	freq(Hz),CH1 (mVpp),CH1 (deg),CH2 (mVpp),CH2 (deg),CH3 (mVpp),CH3 (deg),CH4 (mVpp),CH4 (deg),CH5 (mVpp),CH5 (deg)										
2	1,4005.295166	0.200744629	4013.850098	0.20275879	3994.478027	0.202575684	3998.105469	0.196716309	4022.166748	0.200164795	
3	2,4005.304932	-0.263480823	4013.858887	-0.279544055	3994.469727	-0.280020326	3998.112793	-0.291781577	4022.171387	-0.284743875	
4	3,4005.252441	-0.770359992	4013.80957	-0.764448523	3994.423096	-0.765165686	3998.050781	-0.782716929	4022.115723	-0.772265613	
5	4,4005.202393	-1.257233262	4013.756348	-1.249355316	3994.369629	-1.250301003	3997.98999	-1.273700118	4022.052246	-1.259794831	
6	5,4005.121338	-1.744756698	4013.685791	-1.73493123	3994.298096	-1.736122488	3997.907715	-1.765354514	4021.982178	-1.747892379	
7	6,4005.130859	-2.232411861	4013.703369	-2.22061944	3994.312256	-2.222016003	3997.902832	-2.257156846	4021.982178	-2.236232757	
8	7,4005.070801	-2.720140457	4013.667969	-2.706326246	3994.272949	-2.707989931	3997.817139	-2.748962402	4021.922119	-2.724522352	
9	8,4005.023926	-3.208982944	4013.629395	-3.193228483	3994.238281	-3.195092678	3997.780029	-3.241923809	4021.873535	-3.214062213	
10	9,4004.976318	-3.696049451	4013.580322	-3.678344011	3994.200195	-3.680470228	3997.702637	-3.73310852	4021.818604	-3.701734066	
11	10,4004.933838	-4.184597969	4013.563232	-4.16493988	3994.159668	-4.167294979	3997.628662	-4.225831508	4021.762939	-4.19094181	
12	15,4004.873779	-6.62865591	4013.582275	-6.599187374	3994.17041	-6.602730751	3997.359375	-6.69031763	4021.643555	-6.638111114	
13	20,4004.751221	-9.080265998	4013.537842	-9.040908813	3994.139404	-9.045735359	3996.933594	-9.162135124	4021.44043	-9.092760086	
14	25,4004.466064	-11.54249096	4013.394775	-11.4934206	3993.979736	-11.49941254	3996.276855	-11.64457417	4021.074707	-11.55797005	
15	30,4003.873291	-14.0155735	4012.9729	-13.95675468	3993.539795	-13.96399498	3995.221924	-14.13750935	4020.354736	-14.03395844	
16	35,4003.088379	-16.50226021	4012.38623	-16.43383217	3992.924805	-16.44230652	3993.875977	-16.64371872	4019.412109	-16.52345657	
17	40,4001.740479	-19.00686073	4011.272949	-18.92885208	3991.781494	-18.93855667	3991.887451	-19.16742897	4017.890869	-19.03066635	

How to read dat files of ELOG-DUAL/ELOG1K

ELOGDUAL_BINARY

ELOGDUAL_CAL

0 → 0 (ELOG1K), 1 (ELOG-DUAL ADU-mode), or 2 (ELOG-DUAL PHX-mode)

../../../../Calib/ELOG_No6.txt ELOG-DUAL/ELOG1K calibration file with its path. The file format is shown in the next slide.

0.0

ELOGMT_BINARY

ELOGMT_CAL

0 → 0 (ADU mode) or 1 (PHX mode)

../../../../Calib/ELOG-MT_SN004042.txt → ELOG-MT calibration file with its path.

0.0

ELOGMT_READ_OPTION

7

LOGGER_CAL_DIRECTORY

../../../../Calib

(continue to the next slide)

→ A real number used to correct the time difference between loggers.
When all data are measured with ELOG, one can write 0.0 because no correction is required.

→ To read the magnetic field (H_x , H_y , H_{rx} , and H_{ry}) from ELOG-MT dat files, 7 is written under this option.

→ Under this directory, the filter coefficient files of ELOG-DUAL/ELOG1K and ELOG-MT should be located.

- ELOG1K: firh.txt, firf.txt
- ADU mode of ELOG-MT and ELOG-DUAL: firh_adu.txt, firf_adu.txt
- PHX mode of ELOG-MT and ELOG-DUAL: firh_phx.txt, firf_phx.txt, firf_phx.txt

How to read .dat files of ELOG-DUAL/ELOG1K

DATA_FILES

115200

../Site2_ELOG1K/20221023/20221023-010000_32Hz.dat

0

../Site2_ELOG1K /20221023/20221023-010000_32Hz.dat

0

../Site1_ELOGMT/20221023/20221023-010000_32Hz.dat

0

../Site1_ELOGMT/20221023/20221023-010000_32Hz.dat

0

../Site3_ELOGMT/20221023/20221023-010000_32Hz.dat

0

../Site3_ELOGMT/20221023/20221023-010000_32Hz.dat

0

TRACMT can read .dat file of ELOG-DUAL/ELOG1K when the keyword “ELOGDUAL_BINARY” is written.

TRACMT can read .dat file of ELOG-MT when the keyword “ELOGMT_BINARY” is written.

Because “ELOGMT_READ_OPTION” is 7 (H_x , H_y , H_{rx} , H_{ry}), the time-series of the magnetic field are read from the ELOG-MT dat files.

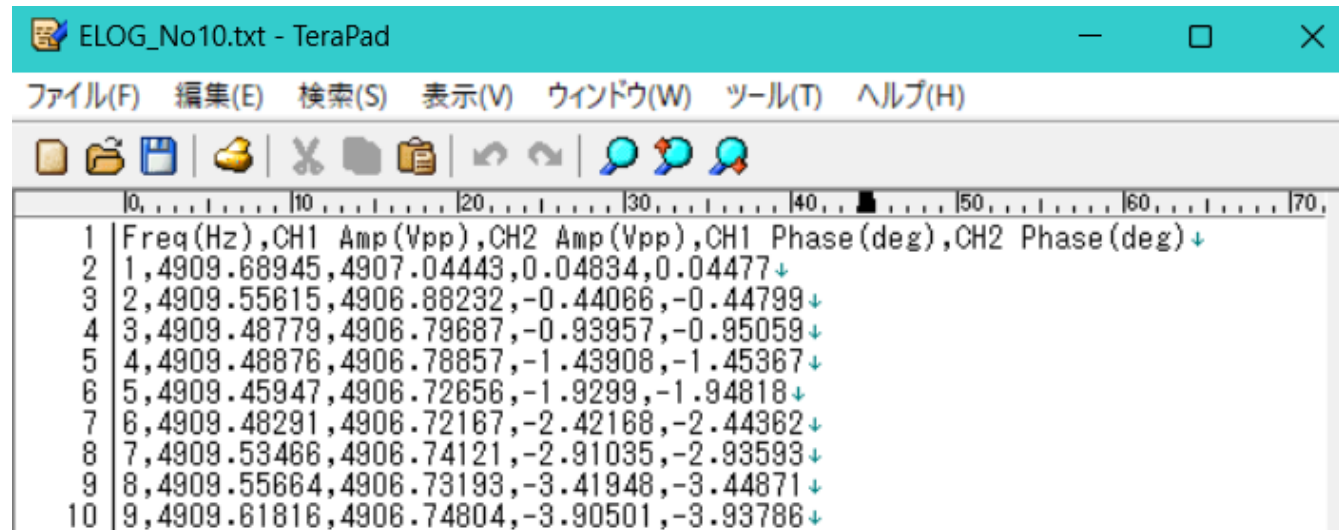
If your compiler supports C++17, you can use the following function by activating the preprocessor “_USE_FILESYSTEM” when compiling.

If you write “*.dat” under the directory name, all data files (of the corresponding sampling frequency) under the directory are automatically read.

The file format of the ELOG-DUAL/ELOG1K calibration file

- The first line is the comment line.
- Below the first line, frequency-dependent characteristics of each channel should be written, comma delimited.
- The first column is the frequency (Hz). The second and third columns are the amplitudes of Ch#1 (E_x) and Ch#2 (E_y), respectively. The fourth and fifth columns are the phase (deg.) of Ch#1 (E_x) and Ch#2 (E_y), respectively.

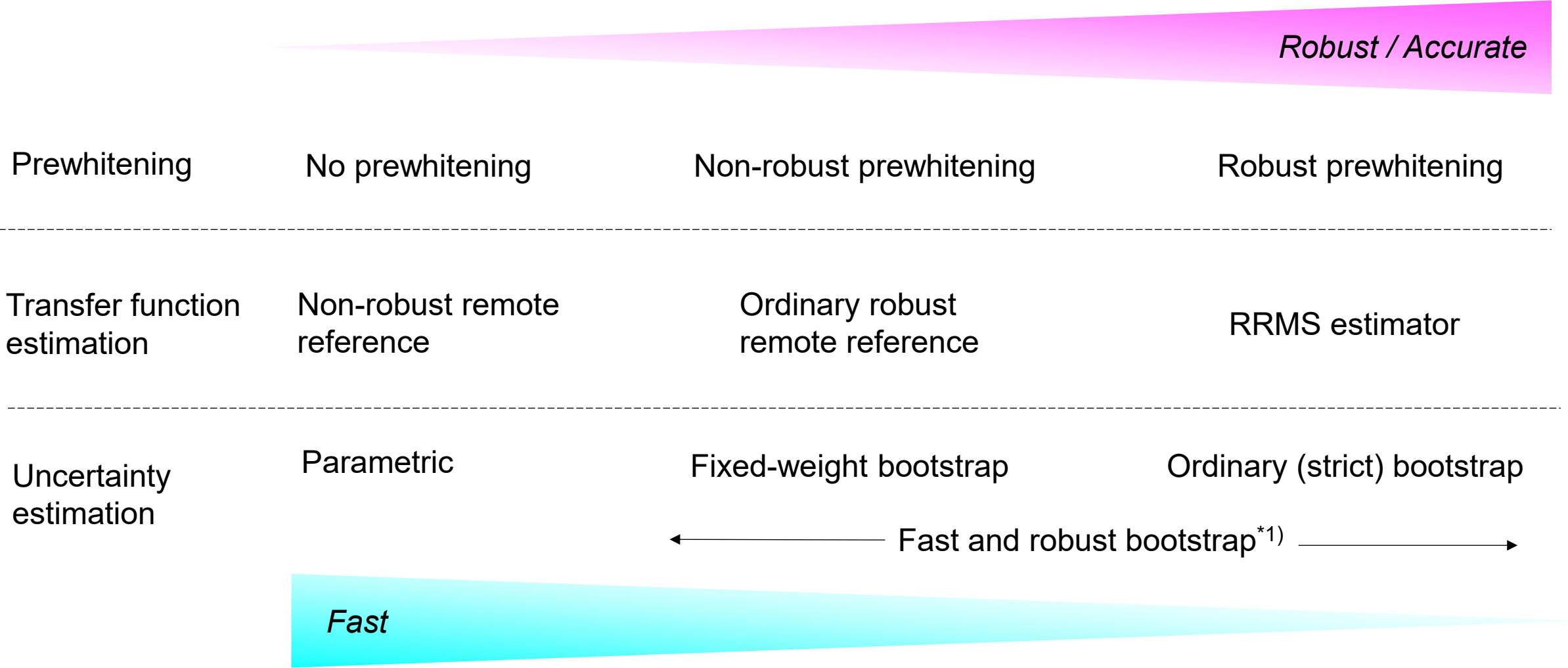
Example of the ELOG-DUAL/ELOG1K calibration file



1	Freq(Hz),CH1 Amp(Vpp),CH2 Amp(Vpp),CH1 Phase(deg),CH2 Phase(deg)
2	1,4909.68945,4907.04443,0.04834,0.04477
3	2,4909.55615,4906.88232,-0.44066,-0.44799
4	3,4909.48779,4906.79687,-0.93957,-0.95059
5	4,4909.48876,4906.78857,-1.43908,-1.45367
6	5,4909.45947,4906.72656,-1.9299,-1.94818
7	6,4909.48291,4906.72167,-2.42168,-2.44362
8	7,4909.53466,4906.74121,-2.91035,-2.93593
9	8,4909.55664,4906.73193,-3.41948,-3.44871
10	9,4909.61816,4906.74804,-3.90501,-3.93786

Tradeoff between robustness and calculation speed

There are tradeoffs between robustness/accuracy and calculation speed



^{*1)} The fast and robust bootstrap method can provide standard errors comparable to those obtained by the ordinary bootstrap method as fast as the fixed-weight bootstrap method unless the uncertainty of the transfer function is large.

Example of parameter setting (1) (Fast but not robust case)

PROCEDURE		
0	}	Non-robust remote reference
MESTIMATORS		
-1		
-1	}	No prewhitening
ERROR_ESTIMATION		
0		Parametric error estimation

Example of parameter setting (2) (Intermediate case)

PROCEDURE		
0	}	Ordinary robust remote reference using the M-estimator
MESTIMATORS		
0		
1	}	Non-robust prewhitening
PREWHITENING		
0		
10	}	Fixed-weight bootstrap
5		
ERROR_ESTIMATION		
1		

Example of parameter setting (3) (robust but not efficient case)

PROCEDURE		RRMS estimator
1		
PREWHITENING		
1	}	Robust prewhitening
10		
5		
ERROR_ESTIMATION		
3		Fast and robust bootstrap